Discussion Exercise 9

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1. Work

For each situation:

- Draw a before-and-after pictorial representation
- Draw and label the displacement vector $\vec{r}$ on your diagram
- Draw a free-body diagram showing *all* forces acting on the object.
- Make a table beside your diagrams showing the sign (+,-, or 0) (i) of the work done by each force seen in your free-body diagram, (ii) of the net work W_{net}, and (iii) of K, the change in the kinetic energy of the object.

1). An elevator moving upward at a constant speed.

2). A descending elevator brakes to a halt.

1. Work

3). A box slides down a frictionless slope.

4). A ball is thrown straight up. Consider the ball from one microsecond after it leaves your hand until the highest part of the trajectory of the ball.

5). A ball is thrown straight up. Consider the ball from the instant you begin moving your hand until you release the ball.

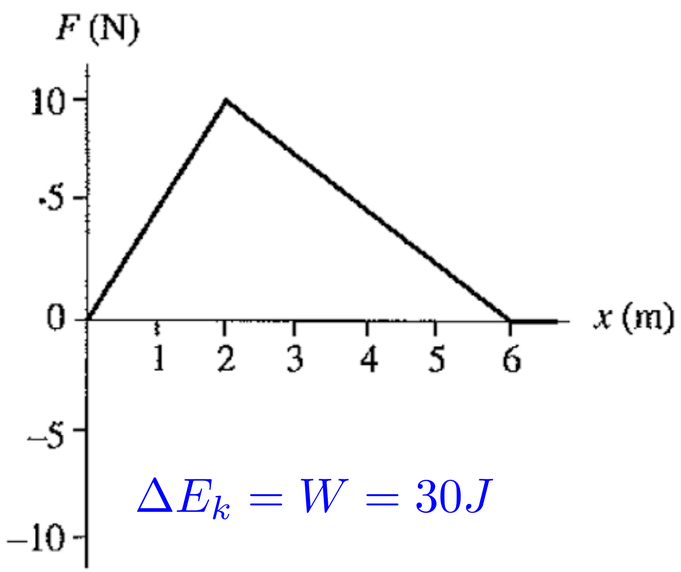
2. Work done by Variable Forces

A particle moving along the x-axis experiences the forces shown below. How much work does each force do on the particle? What is each particle's change in kinetic energy?

Definition: $W = \int_{x_i}^{x_f} F dx$

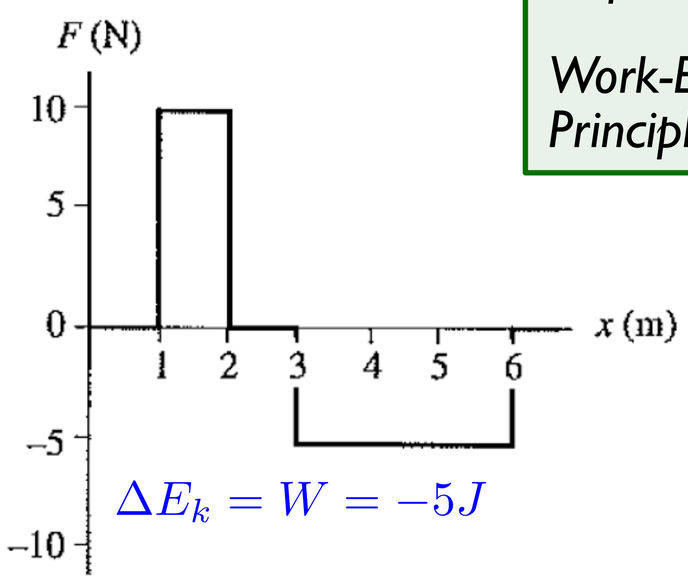
Work-Energy Principle: $W = \Delta E_k$

a.



$\Delta E_k = W = 30J$

b.

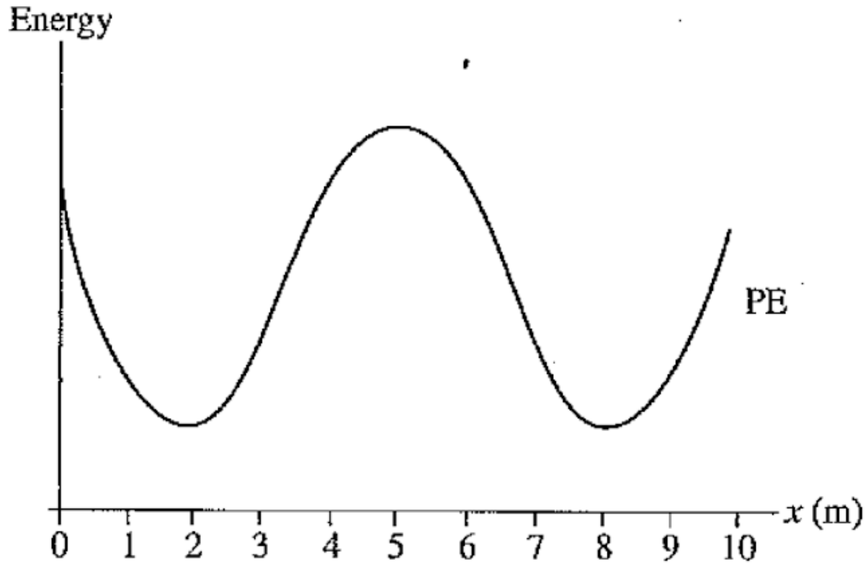


$\Delta E_k = W = -5J$

By definition, the work is the area under the F-x curve.
The change in KE is the work down by the external force according to Work-Energy Principle.

3. Force and Potential Energy

The graph shows the potential-energy curve of a particle moving along the x-axis under the influence of a conservative force.



a. In which intervals of x is the force on the particle to the right?

From 0 to 2m and from 5m to 8m (where the slope is negative).

b. In which intervals of x is the force on the particle to the left?

From 2m to 5m and from 8m to 10m (where the slope is negative).

c. At what value or values of x is the magnitude of the force a maximum?

At x=0, x=3.5m, x=6.5m, and x=10m (The magnitude of a force is a maximum where the slope has its greater magnitude)

$$F_x = -\frac{dV}{dx}$$

The conservative force is negative of the slope of V-x curve

4. Conservation of Energy

$$KE = \frac{1}{2}mv^2$$

Give a *specific example* of a situation (and identify the objects in the system) in which:

- $W_{ext} \rightarrow K$ with $\Delta U = 0$ and $\Delta E_{th} = 0$ — *Push a box across a frictionless surface*
- $W_{ext} \rightarrow U$ with $\Delta K = 0$ and $\Delta E_{th} = 0$ — *Lift a box at a constant speed*
- $K \rightarrow U$ with $W_{ext} = 0$ and $\Delta E_{th} = 0$ — *A ball thrown upward until reaching its peak*
- $W_{ext} \rightarrow E_{th}$ with $\Delta K = 0$ and $\Delta U = 0$ — *Push a box at constant speed across a surface with friction*
- $U \rightarrow E_{th}$ with $\Delta K = 0$ and $W_{ext} = 0$ — *A box slides down a ramp with friction at constant speed*

W_{ext}: work down by external force
K: kinetic energy
U: potential energy
E_{th}: thermal energy

$W_{ext} = 0$: Work down by external force is zero
$\Delta K = 0$: Speed remains the same
$\Delta U = 0$: Height remains the same
$E_{th} = 0$: No friction